Evaluating Rating Reliability: A Cross-Platform Analysis of Movie Ratings and Review Sentiment

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Introduction

Background

Ratings differ across platforms influenced by users, systems, and environments
Are these ratings reliable, and do they reflect audience perception consistently?

Research Focus

Compare consistency of ratings across platforms
Examine alignment between ratings and review sentiment
Analyze patterns of user rating behavior

Approach

1. Compare TMDb and IMDb ratings
2. Analyze user behavior with MovieLens
3. Apply sentiment analysis on TMDb reviews

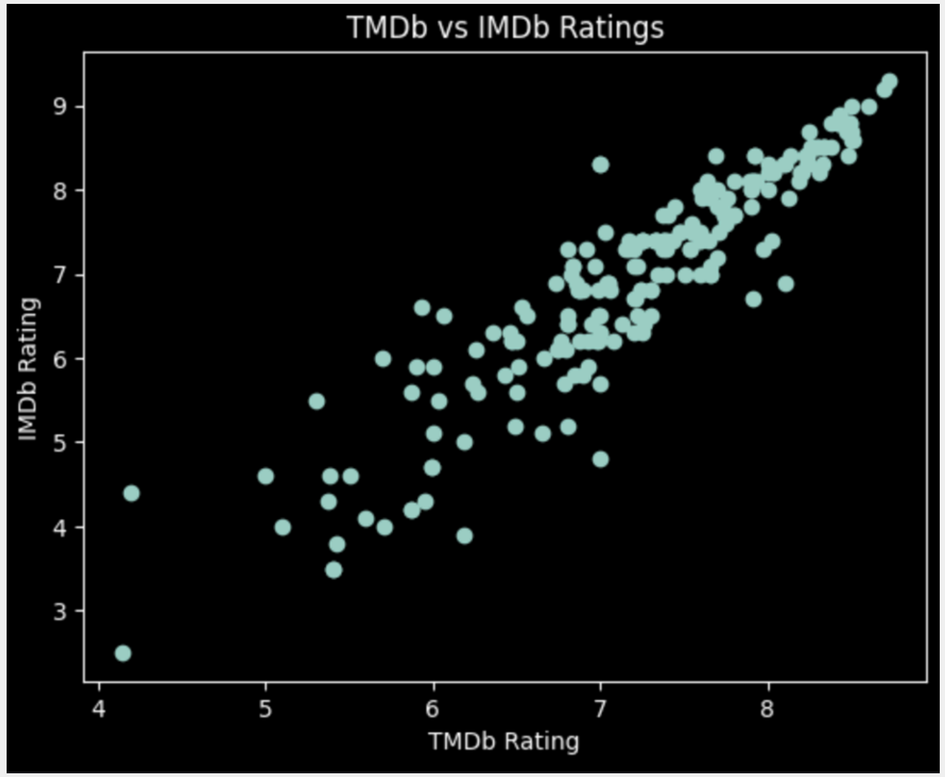
This project provides a practical perspective on how platform ratings reflect user perception and how they should be interpreted in decision-making.

Data sources

Source	Type	Source URL	List of Fields	Format	Purpose	data size
TMDb	API	https://www.themoviedb.org/	ID, Title, Vote Average, Vote Count, Release Date, Genres	json	Platform ratings + sentiment	300+
IMDb	Dataset	https://developer.imdb.com/non-commercial-datasets/	Movie ID, Title, Rating, Votes, Release Year	tsv	Platform comparison	40,000+
MovieLens	Dataset	https://grouplens.org/datasets/movielens/1m/	Movie ID, Title, Rating, Timestamp	csv	User behavior analysis	3,000+

Summary of the Results

1. Consistency of Ratings Across Platforms
(TMDb vs IMDb)



TMDb vs IMDb Ratings

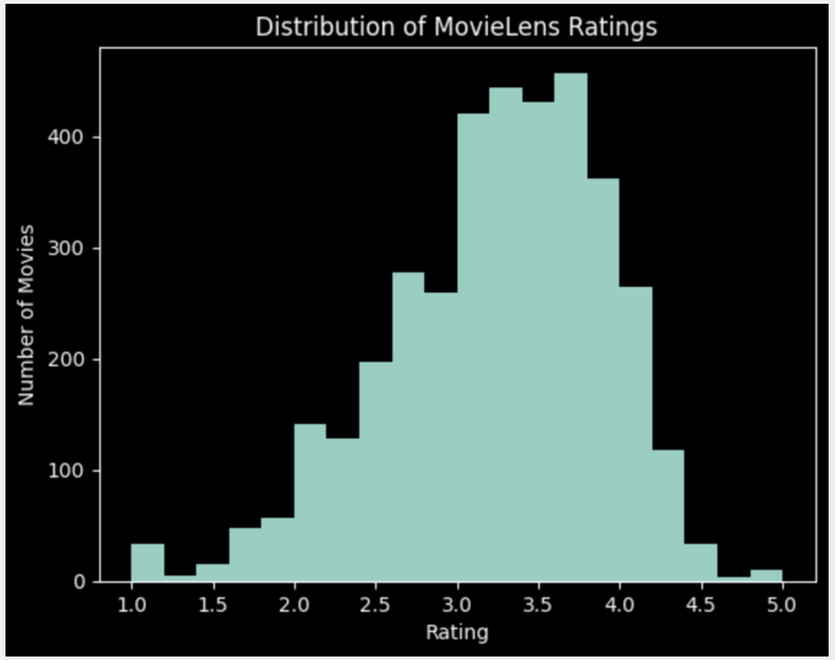
Correlation = 0.92

- Strong positive relationship between TMDb and IMDb ratings
- Higher consistency for highly rated movies
- Greater variation for lower-rated movies
- Overall differences are small

Summary of the Results

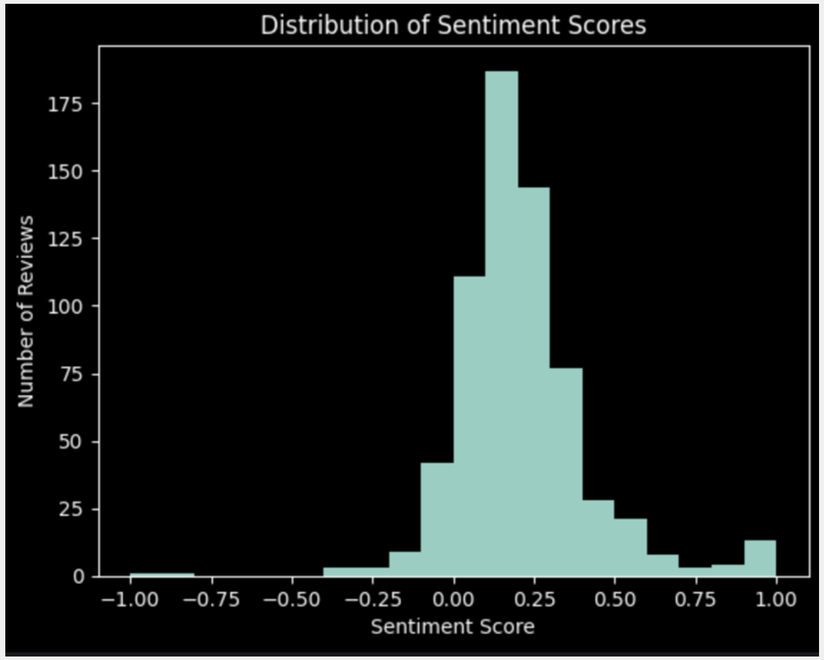
2. User rating behavior and sentiment tendency
(MovieLens + TMDb sentiment)

MovieLens rating distribution



- Ratings concentrated between 3 and 4
- Users tend to give moderate scores
- Popularity does not necessarily lead to higher ratings

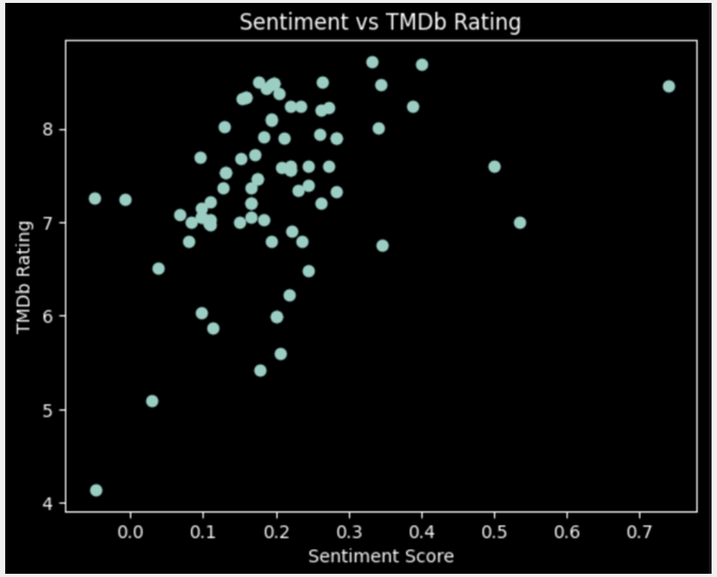
Sentiment distribution



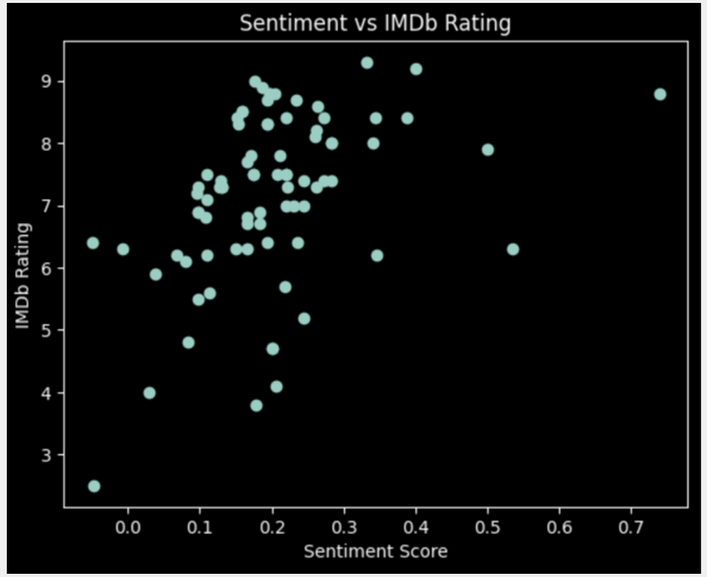
- Most sentiment scores are positive (>0)
- Reviews are generally favorable but not extreme

Summary of the Results

3. Alignment Between Sentiment and Platform Ratings



Sentiment vs TMDb



Sentiment vs IMDb

- General positive trend observed
- Higher sentiment tends to correspond to higher ratings

Limitations

- Relationship is not strongly linear
- Sentiment scores show limited variation

Challenges

- Data accessibility limitations required adjustments in data source selection.
- Applying sentiment analysis and interpreting its results required understanding the limitations of lexicon-based methods.
- Matching movies across platforms was challenging due to inconsistencies in titles and release year formats.